

Sheet 05 — MA-INF 2201 — Computer Vision

1 Multi-Label MRFs

1.1 Alpha Expansion Graph Construction

Legend:

$i \in \{a, b, c, d, e\}$ node index

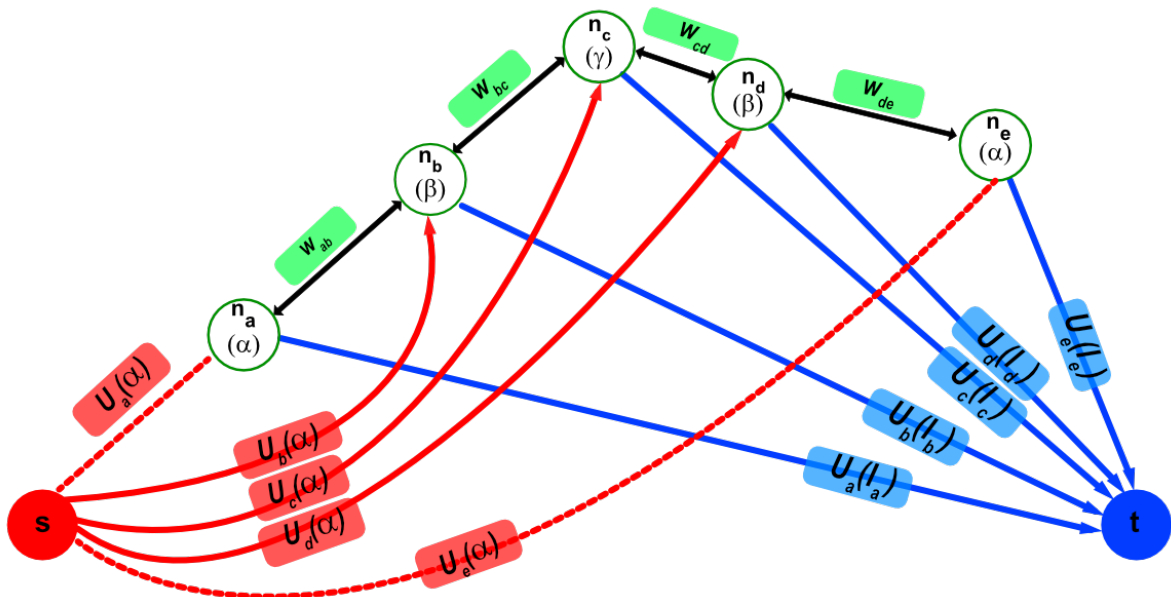
n_i node i in the graph

$s \rightarrow n_i : U_i(\alpha)$ switch node i to label α

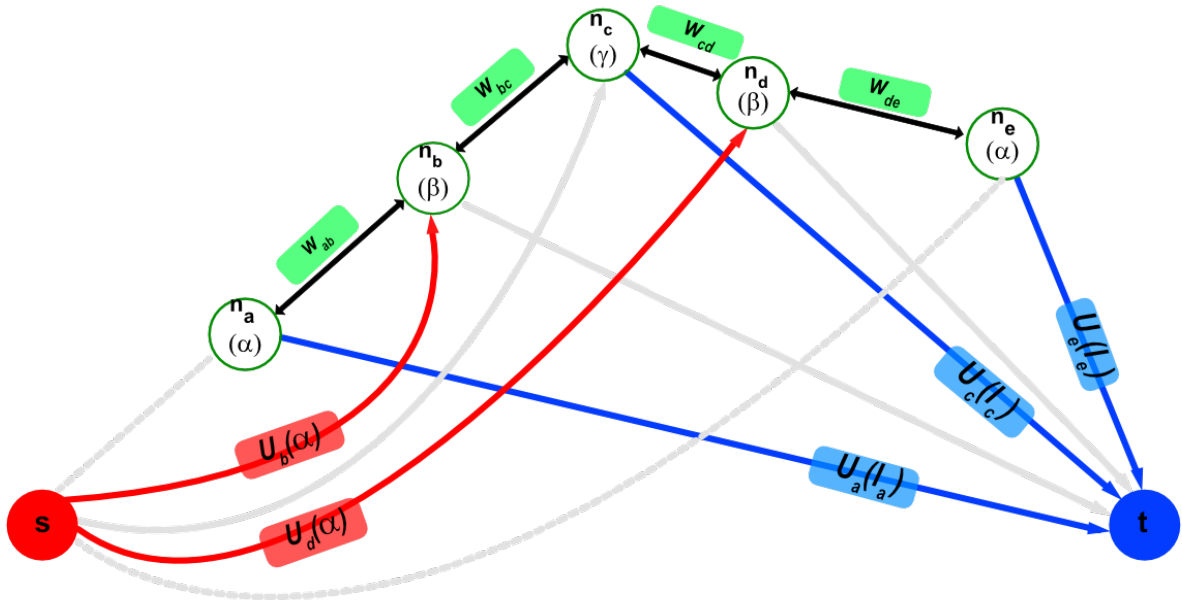
$n_i \rightarrow t : U_i(l_i)$ keep current label l_i of node i

$(n_i, n_j) : w_{ij}$ α -expansion reduced pairwise cost

$$w_{ij} = P_{ij}(l_i, l_j) + P_{ij}(\alpha, \alpha) - P_{ij}(l_i, \alpha) - P_{ij}(\alpha, l_j)$$



1.2 The Minimum Cut



Total Cost of This Configuration

$$C = U_a(l_a) + U_c(l_c) + U_e(l_e) + U_b(\alpha) + U_d(\alpha) + w_{ab} + w_{bc} + w_{cd} + w_{de}$$

CV – MA-INF2201 Sheet 5 Report

Main disadvantages of the core algorithm were that it was limited to the initial scribbles. Meaning, no refinement could be possible, leading to the algorithm not working very well in some of the cases as given below, in which distinguishing between background and foreground was difficult due to complex backgrounds or similar background and foreground.

For the interactive tool, we can see a stark contrast between the results, since we can add scribbles manually again in problematic regions to iteratively improve the background subtraction. For the penguin and the person images, this can be seen very clearly. Adding additional scribbles through the interactive GUI led to much more promising results.

Comparison between Histogram and GMM Color modelling:

Below are the tuning parameters for Histogram and GMM:

1. Histogram

```
tuning_params = [  
    [0.001, 180], # bike - IOU = 0.87  
    [0.003, 90], # aero - IOU = 0.58  
    [0.001, 120], # person7 - IOU = 0.53  
    [0.007, 100], # 208001 - IOU = 0.80  
    [0.009, 90], # scissors - IOU = 0.85  
    [0.035, 150], # 106024 - IOU = 0.45  
]
```

2. GMM:

```
tuning_params = [  
    [0.001, 100], # bike - IOU = 0.88 [n_components = 3 for this]  
    [0.005, 40], # aero - IOU = 0.41 [n_components = 5 for this]  
    [0.003, 120], # person7 - IOU = 0.51 [n_components = 5 for this]  
    [0.5, 50], # 208001 - IOU = 0.35 [n_components = 2 for this]  
    [0.004, 90], # scissors - IOU = 0.85 [n_components = 5 for this]  
    [0.008, 250], # 106024 - IOU = 0.53 [n_components = 3 for this]  
]
```

Histogram Color modelling worked the best overall with a higher average. However, when it came to interactive modelling, GMM was better for complex images. Below are attached images of the results for core vs interactive method for Graph Cut.

